Nitin Nitin

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.02
Jawahar Navodaya Vidyalaya , Mahoba , UP	Class XII (CBSE)	2022	86.20%
Jawahar Navodaya Vidyalaya , Mahoba , UP	Class X (CBSE)	2020	86.40%

SKILLS

- **Languages:** C, C++, JavaScript , TypeScript, SQL
- **Backend & APIs:** Node.js, Express.js, REST APIs, JWT, Zod
- **Databases:** PostgreSQL, Prisma ORM, Redis
- **Testing:** Vitest, Jest, Supertest
- **Tools:** Git, GitHub, Docker, VS Code, Postman, Render, Vercel, k6
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems,Computer Network

PROJECTS

TaskForge -- Project Management Backend

- Built a project management backend using Node.js, Express.js, PostgreSQL and Prisma, supporting workspaces, projects, tasks, members and task assignments.
- Implemented JWT-based authentication and role-based access control (**RBAC**) for workspace members, with authorization checks for protected project and task operations.
- Designed relational database models using Prisma ORM, with migrations, foreign-key relationships and database queries for managing users, workspaces, projects and tasks.
- Added request validation, centralized error handling, **API integration testing and Dockerized development** to improve backend reliability and simplify local setup.

Payment Orchestration & Wallet Platform

- Engineered a modular payment backend supporting **wallet, gateway, and hybrid payment flows**, with merchant/customer authentication, payment orders, and payment-attempt tracking.
- Implemented transactional wallet operations using reserve-release-finalize debit flows, wallet top-ups, transaction history, and automated balance management.
- Secured critical payment and refund operations with **idempotency keys and Prisma transactions**, preventing duplicate processing on retries and rejecting conflicting key reuse.
- Designed split refund workflows across wallet and gateway payments, backed by **Jest/Supertest integration tests, k6 load tests, and Dockerized infrastructure** for correctness and performance validation.

C++ Key-Value Store

- Engineered a modular in-memory key-value store supporting **SET, GET, DELETE, EXISTS, and TTL** expiration through a multithreaded Crow REST API.
- Implemented **O(1) hash-based lookup** and LRU eviction using modular storage and caching layers, enabling efficient key retrieval and automatic removal of least-recently-used entries.
- Added append-only persistence for data recovery and Dockerized the service, separating storage, caching, and persistence layers for a modular and maintainable architecture.

POSITIONS OF RESPONSIBILITY

Volunteer, Office of Career Services (OCS), IIT Hyderabad

- Assisted in coordinating and managing Online Assessments (OAs) and company recruitment processes, ensuring smooth execution of placement activities and timely resolution of operational issues.
- Supported company interviews and candidate coordination, helping maintain seamless communication between recruiters, students and the placement team during recruitment drives.

EXTRACURRICULAR

NSS Volunteer

- Contributed to social welfare initiatives, community outreach drives and public awareness campaigns, supporting organized on-ground activities and promoting community engagement

Volunteer, Fresher Induction & Admission Support

- Assisted with student admission and orientation activities, coordinating documentation, campus tours and query resolution while helping ensure a smooth onboarding experience for incoming students.

RELEVANT COURSES

- **Computer Science & AI -** Introduction to Programming(C language) , Machine Learning and its Applications
- **Mathematics(IITH curriculum) -** Calculus, Linear Algebra, Differential Equations, Probability, Complex Variables
- **Supervised Machine Learning: Regression and Classification -** Coursera
- **Unsupervised Learning, Recommenders, Reinforcement Learning** - Coursera